

Savitzky-Golay optimizer diagnostics

Objective: mean squared translational/angular acceleration residual; angular residual uses the reference inertia/mass metric

Tolerances: {'ftol': None, 'xtol': None, 'gtol': 1e-06, 'note': 'ftol and xtol are disabled by default in the SG estimator; gtol or max_nfev terminates the solve unless explicitly overridden'}

Selected strict-ZOH solve

status=1 success=True nfev=4 njev=4

message='gtol' termination condition is satisfied.

reported optimality=5.663680902046053e-08

initial={'cost': 1.4331464615421048, 'gradient_inf_norm': 0.02053814629663009, 'gradient_l2_norm': 0.027530183015701384, 'jacobian': {'condition_number_nonzero_subspace': 408.0332138472979, 'numerical_rank': 14, 'numerical_tolerance': 1.0479955355370928e-10, 'singular_values': [119.12549983167074, 117.86745831554032, 102.90168037525311, 48.59950510014564, 44.7267682690671, 25.86922556740349, 12.990320334591306, 11.738370224613602, 11.31125282324794, 11.042170408328289, 9.896454229190898, 3.7985597127858926, 3.7926463357207343, 0.2919504976284901]}, 'residual_l2_norm': 1.6930129719184699}

final={'cost': 1.4331463605116408, 'gradient_inf_norm': 5.663680902046053e-08, 'gradient_l2_norm': 6.723654257363316e-08, 'jacobian': {'condition_number_nonzero_subspace': 408.250079270499, 'numerical_rank': 14, 'numerical_tolerance': 1.0479917933840093e-10, 'singular_values': [119.12507446167497, 117.86739539540392, 102.90167023807231, 48.59958083430747, 44.72678717274076, 25.86924212171634, 12.990294848196791, 11.738250921741043, 11.311155210370961, 11.04203328143877, 9.896330947262657, 3.798568511310721, 3.792639650432285, 0.29179436945741505]}, 'residual_l2_norm': 1.6930129122435191}

coordinate_step_l2_norm=0.0002695517511298434

cost_reduction=1.0103046399478899e-07

gradient_tolerance_satisfied=True

finite_difference_max_relative_error=9.312112664126403e-12

finite_difference_directional_checks=[{'analytic_l2_norm': 74.37081263691395, 'difference_l2_norm': 6.925493861999874e-10, 'direction': 'negative_gradient', 'finite_difference_l2_norm': 74.37081263717266, 'relative_error': 9.312112664126403e-12, 'scheme': 'central', 'step': 6.0554544523933395e-06}]

All smoothstep and strict-ZOH stage details are in optimizer_diagnostics.json.